

AJALON PETERSON

ajalon.peterson@gmail.com • 612-505-9646 • ajalonpeterson.com • github.com/ajalonpeterson

EDUCATION

B.S. Geography and Plant Science

September 2022 - May 2026

University of Minnesota

Minneapolis, MN

- Geography (GIS) GPA: 3.72/4.00 | Cumulative GPA: 3.48/4.00
- Dean's List: 3 terms, Green Collar Scholar, UMN Sustainability Corps Alum
- Relevant Coursework: Advanced GIS • Advanced Spatial Data Science • Remote Sensing • Advanced Geocomputing

PROFESSIONAL EXPERIENCE

UNIVERSITY OF MINNESOTA CONTROLLED ENVIRONMENT AGRICULTURE LAB

St. Paul, MN

Undergraduate Research Assistant

June 2024 - June 2025

- Designed and led a 6-month hydroponic experiment to evaluate the performance of a multiphase airlift pump prototype, collecting key data (pH, EC, dissolved oxygen) while refining monitoring protocols for optimal results
- Analyzed experimental datasets in Python, R, and JMP to identify system performance trends and produce reproducible analyses supporting research conclusions

ARGONNE NATIONAL LABORATORY

Minneapolis, MN

Solar Field Ecology Intern

May 2024 - September 2024

- Conducted ecological field surveys at eight utility-scale solar facilities, collecting biodiversity and vegetation data to evaluate the ecological impacts of ecovoltaic development
- Contributed field data and ecological surveys supporting Department of Energy funded research, with contributions acknowledged in two peer-reviewed publications

MIDWEST MUSHROOMS

Lakeville, MN

Production Specialist

December 2021 - August 2022

- Cultivated and harvested 5 varieties of culinary mushrooms, adhering to field mycology procedures and conducting weekly inspections to prevent diseases and contamination
- Managed sales at weekly farmers markets, generating over \$1,000 in daily sales by engaging customers and promoting company values effectively

LEADERSHIP EXPERIENCE

UMN INSTITUTE ON THE ENVIRONMENT

St. Paul, MN

Clean Energy Leader & Sustainability Corps Alumn

January 2024 - December 2024

- Utilized ArcGIS Pro & AGOL to analyze and visualize MPCA contamination data, delivering spatial insights to local nonprofits and community stakeholders regarding development risks near a Superfund site
- Designed a climate-resilient urban forestry plan for the City of Faribault, developing species recommendations across 3 distinct spatial environments to target a 10+ point improvement in tree equity

ACHIEVEMENTS & TECHNICAL SKILLS

Research Recognition: Contributions acknowledged in two peer-reviewed publications:

Szoldatits et al. (2025), Global Ecology and Conservation. <https://doi.org/10.1016/j.gecco.2025.e03864>;

Walston et al. (2025), Journal of Applied Ecology. <https://doi.org/10.1111/1365-2664.70208>

Programming: Python • R • SQL

Geospatial: ArcGIS Pro • AGOL • Google Earth Engine • ArcPy • ESRI Field Maps

Software & Tools: GitHub • VS Code • Jupyter Notebooks • JMP

Field Methods: Native/invasive plant ID • Transect-based vegetation sampling • Water quality monitoring (pH, EC, DO)